Class activity part 2

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

In a study of patients with tumors, scientists were interesting in determining the relationship between the size of tumors in centimeters (X) found on lymph nodes and whether or not the tumor was cancerous (Y). Let $Y_i = 1$ if patient i in the study has a tumor that is cancerous, and $Y_i = 0$ if the tumor is not cancerous. Let $\pi_i = P(Y_i = 1)$.

The data from the first five patients in the study are:

Tumor cancerous	Yes	No	No	Yes	No
Size of tumor (cm)	6	1	0.5	4	1.2

Part II

A second scientist (Scientist B) proposes that the probability π_i *does* depend on tumor size, and suggests the logistic regression model

$Y_i \sim Bernoulli(\pi_i)$

$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \ Size_i$

3. Scientist B suggests $\hat{\beta}_0 = -2$ and $\hat{\beta}_1 = 0.5$. Calculate $\hat{\pi}_i$ for each of the five patients in the data above.

4. Using your probability estimates from Exercise 3, calculate the likelihood $L(\hat{\beta}_0, \hat{\beta}_1) = P(y_1, ..., y_5 | \hat{\beta}_0, \hat{\beta}_1)$.